**1.** Després d'una darrera anàlisi del sistema de salut, s'ha analitzat l'esquema de bases dades actual i ens demanen introduir les modificacions següents a les definicions de les taules:

**a)** Es demana donar les sentències SQL per a la creació de la taula VISITA (VISIT).

**b)** El número de pacient, Pat_Number, és un nombre entre 10.000 i 10.000.000.

**c)** És vol suprimir de la taula PACIENT (PATIENT) la darrera visita (columna Last_Visit), atès que es pot obtenir d'altres formes.

**d)** Donada la dependència de les receptes respecte de les visites, es vol que a la taula RECEPTA (PRESCRIPTION) només hi hagi una restricció de clau forana formada per la tripleta d'atributs número de col·legiat del metge (Doc_Number), el número de pacient (Pat_Number) i la data de la visita (Visit_Date) de VISITA (VISIT) en comptes de les dues restriccions de clau forana actuals.

**e)** A la taula METGE (DOCTOR), es vol incorporar el número d'identificació nacional (DNI) que és únic per a cada metge però pot ser nul si no se'n disposa.

```
CREATE TABLE DOCTOR
  (
    Doc_Number INTEGER,
    Name        VARCHAR(50) NOT NULL,
    Specialty   VARCHAR(50) NOT NULL,
    Address     VARCHAR(50) NOT NULL,
    City        VARCHAR(30) NOT NULL,
    Phone       VARCHAR(10) NOT NULL,
    Salary      DECIMAL(8,2) NOT NULL,
    DNI         VARCHAR(10) UNIQUE,
    CONSTRAINT pk_DOCTOR PRIMARY KEY (Doc_Number),
    CONSTRAINT ck_Salary CHECK (Salary > 0)
  );
```

Output pane

Data Output | Explain | Messages | History

```
NOTICE:  schema "ubd_20151" does not exist, skipping
Query returned successfully with no result in 74 msec.
```

```sql
CREATE TABLE PATIENT
  (
    Pat_Number  INTEGER,
    Name        VARCHAR(50) NOT NULL,
    Address     VARCHAR(50) NOT NULL,
    City        VARCHAR(30) NOT NULL,
    DNI                         VARCHAR(10) UNIQUE,
    CONSTRAINT pk_PATIENT PRIMARY KEY (Pat_Number),
    CONSTRAINT ck_Pat_Number CHECK (Pat_Number >= 10000 AND Pat_Number <= 10000000)
    );

CREATE TABLE VISIT
```

Output pane

Data Output | Explain | Messages | History

```
NOTICE:  schema "ubd_20151" does not exist, skipping
Query returned successfully with no result in 74 msec.
```

```sql
CREATE TABLE PRESCRIPTION
  (
    Presc_Number    INTEGER,
    Drug             VARCHAR(50) NOT NULL,
    Doc_Number        INTEGER NOT NULL,
    Pat_Number        INTEGER NOT NULL,
    Visit_Date        DATE NOT NULL,
    CONSTRAINT PRESCRIPTION_pk PRIMARY KEY (Presc_Number),
    CONSTRAINT PRESCRIPTION_VISIT_fk FOREIGN KEY (Doc_Number, Pat_Number, Visit_Date)
    CONSTRAINT ck_Pat_Number CHECK (Pat_Number >= 10000 AND Pat_Number <= 10000000)
  );
```

Output pane

Data Output | Explain | Messages | History

```
NOTICE:  schema "ubd_20151" does not exist, skipping
Query returned successfully with no result in 74 msec.
```

```sql
CREATE TABLE VISIT
  (
    Doc_Number        INTEGER NOT NULL,
    Pat_Number        INTEGER NOT NULL,
    Visit_Date        DATE NOT NULL,
    Price       DECIMAL(8,2) NOT NULL,
    CONSTRAINT pk_VISIT PRIMARY KEY (Doc_number, Pat_Number, Visit_Date),
    CONSTRAINT VISIT_DOCTOR_fk FOREIGN KEY (Doc_Number) REFERENCES DOCTOR(Doc_Number)
    CONSTRAINT VISIT_PATIENT_fk FOREIGN KEY (Pat_Number) REFERENCES PATIENT(Pat_Numbe
    CONSTRAINT ck_Price CHECK (Price > 0),
    CONSTRAINT ck_Pat_Number CHECK (Pat_Number >= 10000 AND Pat_Number <= 10000000)
    );
```

Output pane

Data Output | Explain | Messages | History

```
NOTICE:  schema "ubd_20151" does not exist, skipping
Query returned successfully with no result in 74 msec.
```

```
SET search_path TO ubd_20151;

SET datestyle = DMY;

BEGIN WORK;
SET TRANSACTION READ WRITE;

--- DOCTOR
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(2
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(2
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(3
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(3
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(9
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(1
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(2
INSERT INTO DOCTOR (Doc_number,Name,Specialty,Address,City,Phone,Salary,DNI) values(3
```

Output pane

Data Output | Explain | Messages | History

```
Query returned successfully with no result in 343 msec.
```

**2. a)** Dissenyar una consulta que permeti obtenir el nom i DNI dels pacients de Barcelona, a qui no s'hagi fet cap recepta des del 2010 (inclòs), ordenats alfabèticament pel nom del pacient.

SELECT name, dni FROM patient WHERE city = ‘Barcelona’
AND pat_number NOT IN (SELECT pat_number FROM prescription
WHERE visit_date >= ‘2010-01-01’) ORDER BY name ASC

```
SELECT name, dni FROM patient WHERE city = 'Barcelona' AND pat_number NOT IN
(SELECT pat_number FROM prescription WHERE visit_date >= '2010-01-01')
ORDER BY name ASC
```

Output pane

Data Output | Explain | Messages | History

| | name character varying(50) | dni character varying(10) |
|---|---|---|
| 1 | Sra. Roig | 911921624Y |
| 2 | Sra Sala | 669133354N |
| 3 | Sr. Pérez | 939842469R |
| 4 | Sr. Sabater | 546372551F |

**b)** Dissenyar una consulta que permeti obtenir els pacients a qui els havien receptat més de cinc medicaments des de l'any 2010. Concretament es demana el nom del pacient, el número de pacient, el nombre de medicaments receptats i la ciutat on viu ordenats pel nombre de medicaments de més gran a més petit.

SELECT name, patient.pat_number, COUNT(prescription.pat_number), city
FROM patient, prescription WHERE patient.pat_number = prescription.pat_number
AND visit_date >= ‘2010-01-01’ GROUP BY name, patient.pat_number, city
HAVING COUNT(prescription.pat_number) > 5
ORDER BY COUNT(prescription.pat_number ) DESC

```
SELECT name, patient.pat_number, COUNT(prescription.pat_number), city
FROM patient, prescription WHERE patient.pat_number = prescription.pat_number
AND visit_date >= '2010-01-01' GROUP BY name, patient.pat_number, city
HAVING COUNT(prescription.pat_number) > 5
ORDER BY COUNT(prescription.pat_number) DESC
```

Output pane

Data Output | Explain | Messages | History

| | name character varying(50) | pat_number integer | count bigint | city character varying(30) |
|---|---|---|---|---|
| 1 | Sra. Alemany | 6574405 | 9 | Barcelona |
| 2 | Sr. Armengol | 9934166 | 6 | Barcelona |

**c)** Crear una vista, anomenada top_patients, la qual reculli el pacient de cada ciutat que ha gastat més en visites (al llarg de tot el període). Concretament es demana una llista dels pacients, ordenada per la despesa on hi consti el nom, el DNI, la ciutat i la despesa acumulada. Busca la informació necessaria per crear una vista a postgreSQL.

CREATE VIEW top_patients AS (SELECT name, dni, patient.city, "Preumax"
FROM patient, (SELECT MAX("Preu") "Preumax", city
FROM (SELECT city, visit.pat_number, SUM(price) "Preu" FROM patient, visit
WHERE visit.pat_number = patient.pat_number GROUP BY city, visit.pat_number)
AS inst GROUP BY city) AS inst1,
(SELECT SUM(price) "Maxqu", visit.pat_number, city FROM visit, patient
WHERE visit.pat_number = patient.pat_number GROUP BY visit.pat_number, city
ORDER BY "Maxqu" DESC) AS inst2 WHERE "Preumax" = "Maxqu"
AND inst1.city = inst2.city AND patient.pat_number = inst2.pat_number)

```
CREATE VIEW top_patients AS (SELECT name, dni, patient.city, "Preumax"
FROM patient, (SELECT MAX("Preu") "Preumax", city
FROM (SELECT city, visit.pat_number, SUM(price) "Preu" FROM patient, visit
WHERE visit.pat_number = patient.pat_number GROUP BY city, visit.pat_number)
AS inst GROUP BY city) AS inst1, (SELECT SUM(price) "Maxqu", visit.pat_number, city
FROM visit, patient WHERE visit.pat_number = patient.pat_number
GROUP BY visit.pat_number, city ORDER BY "Maxqu" DESC) AS inst2
WHERE "Preumax" = "Maxqu" AND inst1.city = inst2.city
AND patient.pat_number = inst2.pat_number)

SELECT * FROM top_patients
```

Output pane

Data Output | Explain | Messages | History

| | name character varying(50) | dni character varying(10) | city character varying(30) | Preumax numeric |
|---|---|---|---|---|
| 1 | Sra. Llach | 673709873Z | Sabadell | 941.00 |
| 2 | Sra. Pons | 227518400P | Tarragona | 949.00 |
| 3 | Sra. Mas | 630733065V | L'Hospitalet | 1234.00 |
| 4 | Sra. Alemany | 176784267B | Barcelona | 2825.00 |
| 5 | Sr. Bosch | 196249494Q | Badalona | 2401.00 |
| 6 | Sr. Nadal | 556260710Y | Girona | 508.00 |

**3.** L'Ajuntament de la ciutat de Vic va decidir subvencionar amb 10 euros per visita en aquells pacients de la localitat que haguessin fet més d'una visita entre els anys 2012 i 2015. Així doncs, s'han de descomptar els 10€ pertinents de cadascuna d'aquestes visites.

Proposeu una única sentència SQL per corregir els registres incorrectes (en cas d'utilitzar-ne més d'una, l'exercici es considerarà incorrecte). Per altra banda, mostreu la relació de registres que s'actualitzen.

**Nota important:** quan proveu d'implementar aquesta actualització, tingueu en compte que sempre heu de partir de la mateixa base de dades, altrament us podeu trobar incoherències en les vostres anàlisis.

UPDATE visit SET price = price-10 WHERE pat_number IN
(SELECT visit.pat_number FROM patient, visit WHERE visit_date >= '2012-01-01'
AND visit_date < '2016-01-01' AND city = 'Vic'
AND patient.pat_number = visit.pat_number GROUP BY visit.pat_number
HAVING COUNT(visit.pat_number) > 1) AND visit_date >= '2012-01-01'
AND visit_date < '2016-01-01'

```
UPDATE visit SET price = price-10 WHERE pat_number IN
(SELECT visit.pat_number FROM patient, visit WHERE visit_date >= '2012-01-01'
AND visit_date < '2016-01-01' AND city = 'Vic'
AND patient.pat_number = visit.pat_number GROUP BY visit.pat_number
HAVING COUNT(visit.pat_number) > 1) AND visit_date >= '2012-01-01'
AND visit_date < '2016-01-01'
```

Output pane

Data Output | Explain | Messages | History

```
Query returned successfully: 8 rows affected, 18 msec execution time.
```